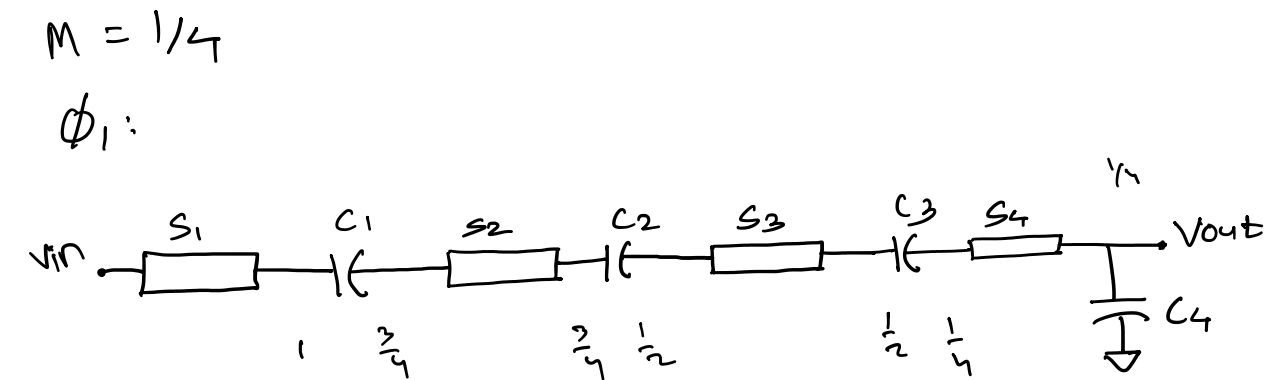
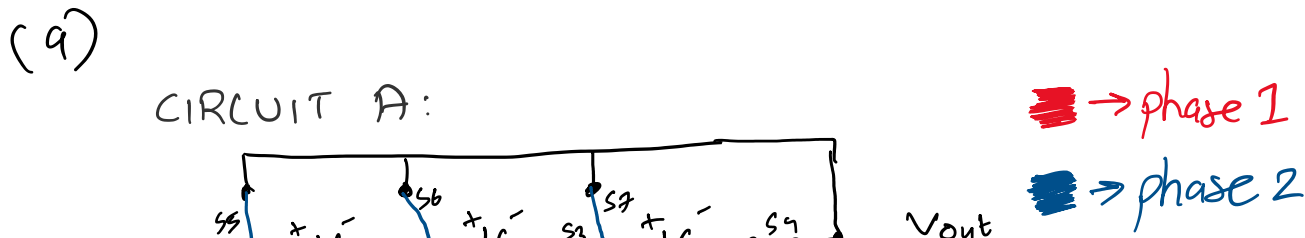
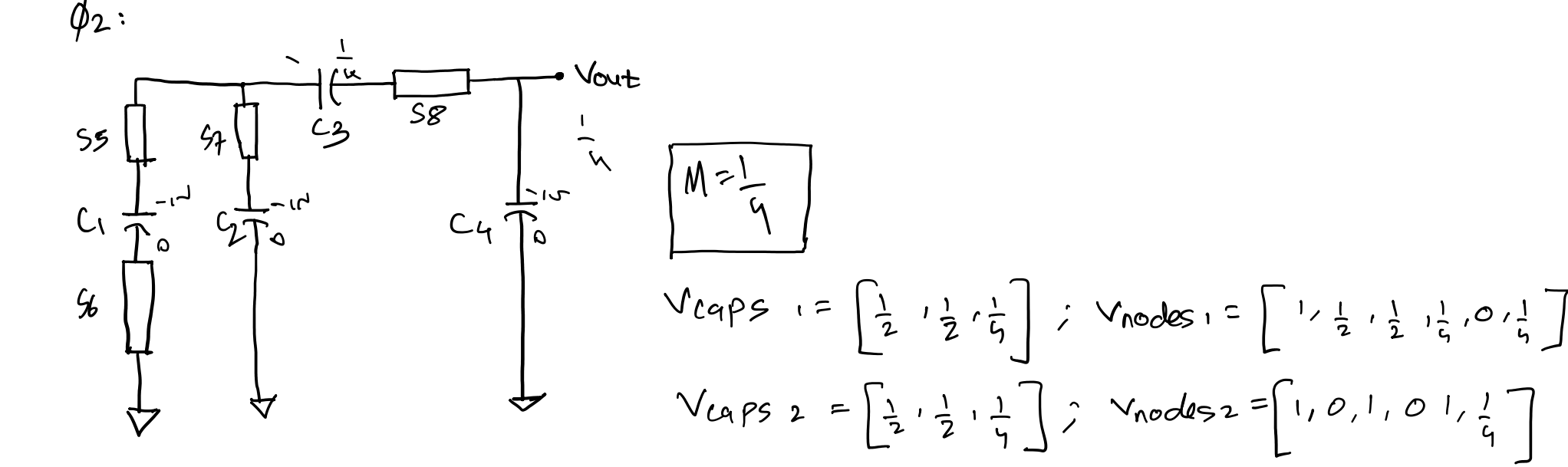
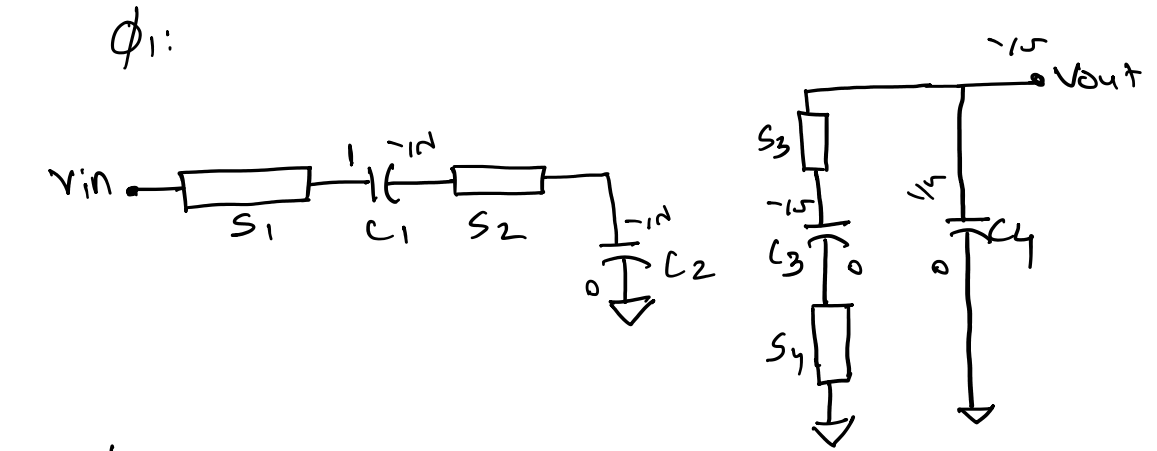
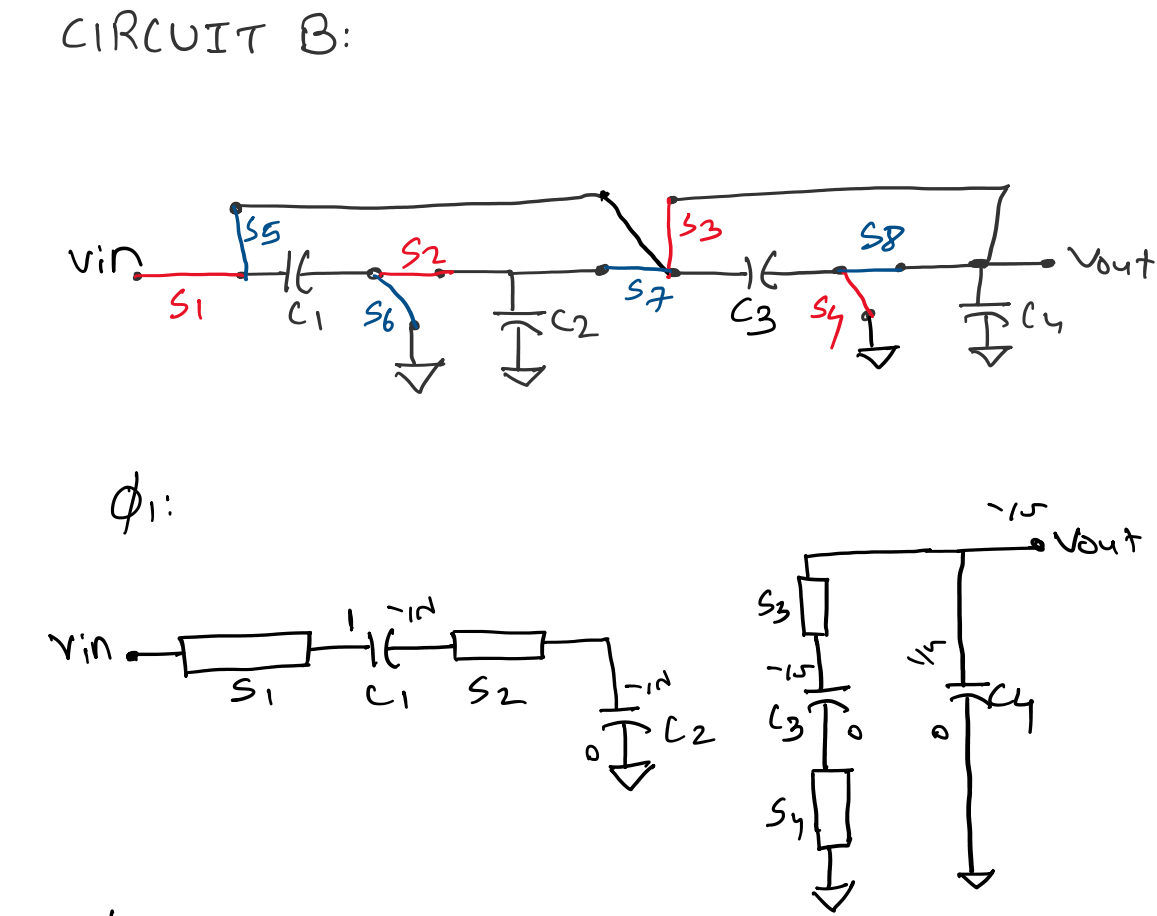


$V_{CAPS1} = \left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right]$; $V_{nodes1} = \left[1, \frac{3}{4}, \frac{3}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4} \right]$

$V_{CAPS2} = \left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right]$; $V_{nodes2} = \left[\frac{1}{4}, 0, \frac{1}{4}, 0, \frac{1}{4}, 0 \right]$

SWITCH	S_1		S_2		S_3		S_4		S_5		S_6		S_7		S_8		S_9		S_{10}			
	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2		
TERM	v_{in}	C_1^+	v_{in}	C_1^+	C_1^-	C_2^+	C_2^-	C_3^+	C_3^-	V_{out}	C_3^-	V_{out}	C_1^+	V_{out}	C_1^+	V_{out}	C_2^+	V_{out}	C_2^+	V_{out}		
VOLT			1	$1/4$		0	$1/4$		0	$1/4$		0	$1/4$		$\frac{3}{4}$	$1/4$		$\frac{1}{2}$	$1/4$		0	
1A V		X	$3/4$		X	$1/4$		X	$1/4$		$3/4$		X	$1/2$		X	$3/4$		X	$1/4$		X

Blocking Voltage = $V_{in} \left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{2}, \frac{1}{4} \right]$



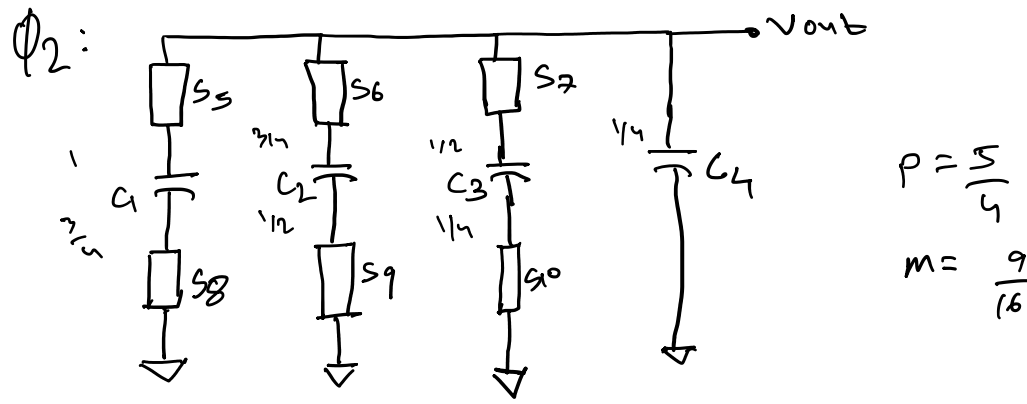
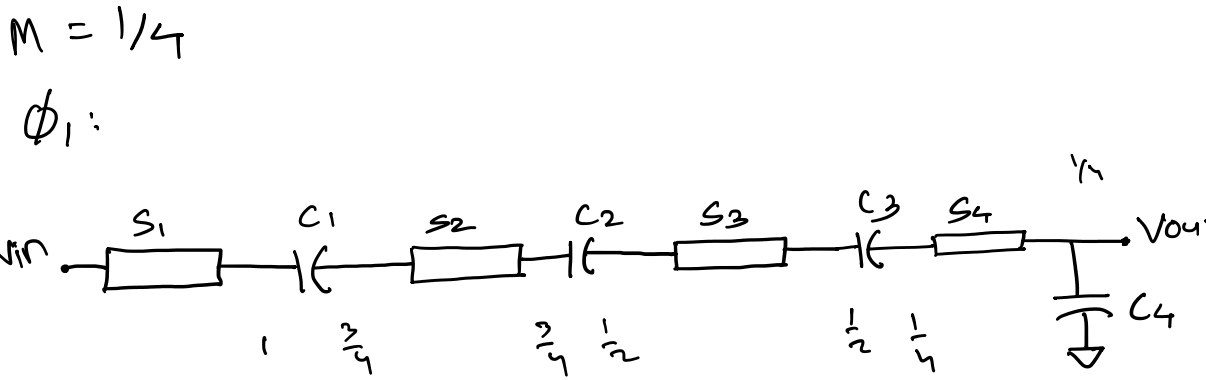
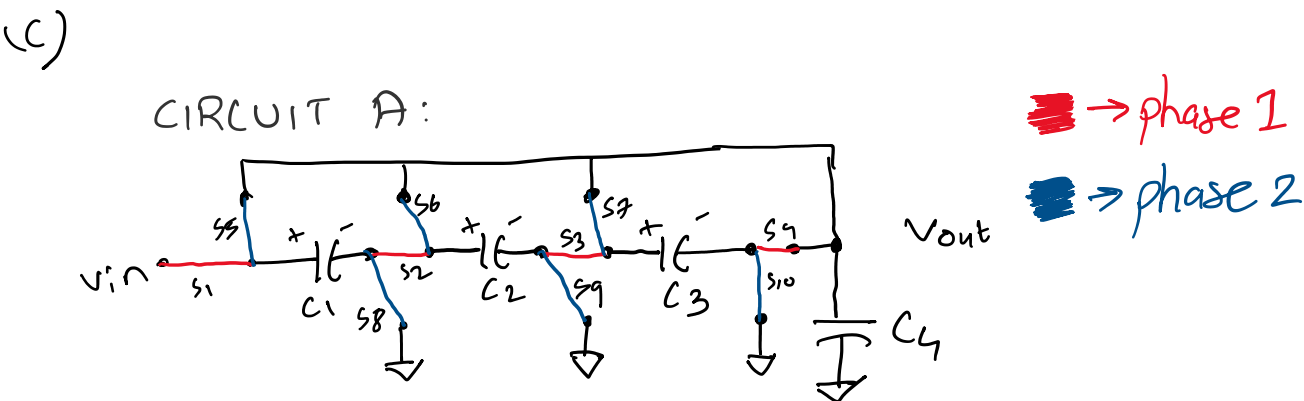
$V_{CAPS1} = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4} \right]$; $V_{nodes1} = \left[1, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, 0, \frac{1}{4} \right]$

$V_{CAPS2} = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4} \right]$; $V_{nodes2} = \left[1, 0, 1, 0, \frac{1}{4}, \frac{1}{4} \right]$

SWITCH	S_1		S_2		S_3		S_4		S_5		S_6		S_7		S_8	
	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2	ϕ_1	ϕ_2
TERM	V_{in}	C_1^+	V_{in}	C_1^+	C_1^-	C_2^+	V_{out}	C_2^-	C_3^+	C_3^+	C_3^+	C_1^+	C_1^-	C_2^-	C_3^+	C_3^+
VOLT		1	$\frac{1}{2}$		0	$\frac{1}{2}$		$\frac{1}{4}$	0		$\frac{1}{4}$	1		$\frac{1}{2}$	0	
1A V	X	$\frac{1}{2}$	X	$\frac{1}{2}$	X	$\frac{1}{4}$	X	$\frac{1}{4}$	$\frac{3}{4}$	X	$\frac{1}{2}$	X	$\frac{1}{4}$	X	$\frac{1}{4}$	X

Blocking Voltages = $V_{in} \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}, \frac{3}{4} \right]$

- (b)
- closer to $V_{in} \rightarrow$ PMOS (High rail)
- closer to $V_{ND} \rightarrow$ NMOS (Low rail)
- Ckt A $\rightarrow$ PMOS $\rightarrow S_1, S_2, \dots, S_7$; NMOS $\rightarrow S_8, S_9, S_{10}$
- Ckt B $\rightarrow$ PMOS $\rightarrow S_1, S_2, S_3, S_5, S_7, S_8$; NMOS $\rightarrow S_4, S_6$



$R_{out} = \sqrt{R_{S2L}^2 + R_{F2L}^2} = \sqrt{\left(\frac{M}{F_S C_T} \right)^2 + (P R_{on})^2}$

but, at f_c , $R_{out} = \sqrt{2} R_{F2L}$

$20 = \sqrt{2} \times \frac{5}{4} \times R_{on}$

$R_{on} = \frac{4 \times 20}{5 \times \sqrt{2}} = \frac{16}{\sqrt{2}} = 11.313 \Omega$

& since at f_c , $R_{eqL} = R_{S2L}$,

$\frac{20}{\sqrt{2}} = \frac{m}{F_S C_T} = \frac{9}{16 \times 10 \times 10^6 \times C_T}$

$C_T = \frac{\sqrt{2} \times 9}{20 \times 16 \times 10} \times MF$

$C_T = 3.977 nF$

Similarly, For circuit B,

$m=1$; $p = \frac{17}{8}$

$\frac{20}{\sqrt{2}} = \frac{17}{8} \times R_{on} \Rightarrow R_{on} = 6.655 \Omega$

$C_T = \frac{\sqrt{2}}{20 \times 10} \times MF = 7.07 nF = C_T$

Now, the area:

The best area distribution:

Circuit A: $C_1 = C_2 = C_3 = 1.33 nF$

In some cycle, the cap has a voltage $V_C = V_{out} = \frac{V_{in}}{4}$

So C_{avg} is more suitable. (for given V_{in}),

so,

Area (floating) = $\frac{3.977}{6} mm^2 = 0.6628 mm^2$

Circuit B:

$C_1 = C_2$ & $C_3 = 2C_1 = 2C_2$

$C_1 = C_2 = 1.77 nF$; $C_3 = 3.54 nF$

all of these caps are still C_{avg}

$\Rightarrow$ Area (floating) = $\frac{7.07}{6} = 1.18 mm^2$

d. $n_{max} = \frac{V_{out}}{m V_{in}} = \frac{0.8}{\frac{1}{4} \times 4} = 0.8$